

— DN HACKS

VESPER

Digital twins that train edge-
autonomous drones for the
comms-denied fight.

RECONSTRUCT • TRAIN • DEPLOY — HUMAN-AUTHORIZED ON EVERY STRIKE

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— THE FRONT LINE

Drones lose the war the moment they lose the link.

In Ukraine, both sides jam the spectrum so hard that GPS and radio simply die. The workaround is desperate — flying drones on **kilometers of fiber-optic cable** that now litter the front.



UKRAINE · EVERY GLINTING THREAD IS ONE DRONE SORTIE'S FIBER, LEFT WHERE IT FELL

A VILLAGE NEAR THE FRONT, SILVERED WITH SPENT FIBER-OPTIC CABLE — DRONES FLOWN BY WIRE BECAUSE RADIO IS DEAD. WAR-REPORTING PHOTO — CREDIT THE SOURCE.

LIVE DOWNLINK

VESPER • TACTICAL

KRAMATORSK • AO 2000 M • N1

T+23.1s

POLICY • SEARCH.PT

RECENTER

SEARCHING

FOUND 1/3

NEUTRALIZED 0/3

ASSETS 16

VESPER

LEAD SPD 9.4 m/s

AGL 86 m

VZ +4.2

AO MAP

KRAMATORSK_MS • 2000 M × 2000 M • N UP

live t=23 s • found 1/3 • reached 0/3 • search.pt

TRACK WAYPOINTS SIGHTED REACHED UNFOUND

drag to pan • scroll to zoom • click a marker

Autonomy fails when the sim isn't the battlefield.

Small, purpose-built models fly the drone itself — search, detect, strike, no link required. Everyone has tried this. It fails because the models train in generic sims that look nothing like where they'll fly. Ours train in the fight.

SIXTEEN AUTONOMOUS DRONES OVER REAL KRAMATORSK TERRAIN — THE WHOLE STACK RUNNING ON THIS MAC.

N

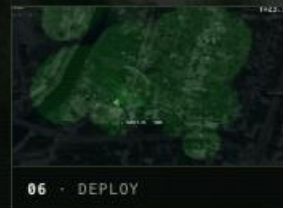
03 / 07

TO NAVIGATE

— THE PIPELINE

Reconstruct, train, **deploy.**

This background is not a photo — it's our sim. Three steps: rebuild the real battlefield in 3D from satellite, lidar, and OpenStreetMap. Fly millions of RL sorties inside it. Ship the weights to the drone — with the RF environment modeled, because **comms is terrain**.



KRAMATORSK — RECONSTRUCTED AUTOMATICALLY. OPERATORS PICK THE LAUNCH POINT AND MARK NO-STRIKE ZONES. OUR DRONES TRAIN, FLY, AND DIE HERE BEFORE THEY EVER FLY FOR REAL.

— INSIDE STEP 04 — TRAINING

Reinforcement learning, millions of sorties in the twin.

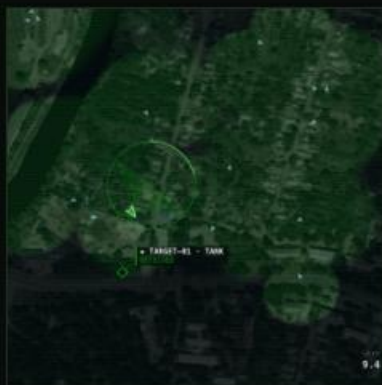


LOST ———> METHODICAL ———> **LETHAL** · SAME POLICY, SAME TERRAIN — THE ONLY THING THAT CHANGED IS TRAINING

Millions of rollouts in the twin — **more flight data than any real fleet could ever log**. Nobody scripts a behavior: reward defines what the operator wants, and the policy improves with every trajectory. What you're about to see live is that policy flying.

● GOING LIVE · KRAMATORSK AO · THIS MAC

Search, detect, report, human-authorized strike.



01

SEARCH

The fleet sweeps the AO autonomously, sharing one map.



02

DETECT

Armor in the treeline. Tagged in seconds.



03

REPORT

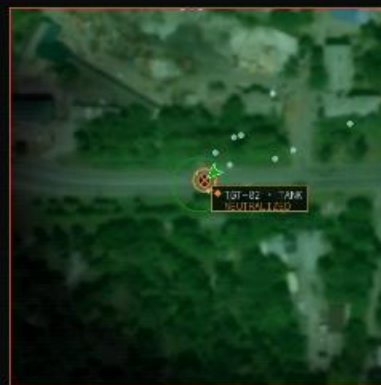
RF model shows where it can reach the operator; it flies there to report.



04

APPROVE

The request hits your screen. You approve — or you don't.



05

STRIKE

On the operator's command, the target is neutralized.

Autonomy without a connection. Fleet coordination without a script. **Nothing fires without a human.**

— VESPER

Deployable at any coordinates **on Earth.**

This is the strike — through the drone's own eyes. Next: environment-specific models for the cities on the front — Vuhledar, Kramatorsk — trained in their twins, deployed in the real world.

EDGE-AUTONOMOUS — RUNS ON A MAC **COMMS**-AWARE, JAM-RESILIENT **HUMAN**-AUTHORIZED, ALWAYS **RL**-TRAINED, NOT SCRIPTED

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The cord is cut.